LAB 7

Polynomial Interpolation

Assignment:

One numerical as suggested by Faculty.

Sample numerical.

Given the data

x	1	2	2.5	3	4	5
$f(x)$	0	5	7	6.5	2	0

Calculate f Lagrange polynomial interpolating polynomials of order 1 through 3. Choose the sequence of the points for your estimates to attain the best possible accuracy. That is, the points should be centered around and as close as possible to the unknown.

Hand Calculation:

For 1st Order

$$f(x) = f(x_0)\frac{(x-x_1)}{(x_0-x_1)} + f(x_1)\frac{(x-x_0)}{(x_1-x_0)}$$

For 2nd Order

$$f(x) = f(x_0)\frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} + f(x_1)\frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)}$$
$$+ f(x_2)\frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}$$

For 3rd Order

$$f(x) = \sum_{i=0}^{3} f(x_i) \prod_{j=0}^{3} \frac{x-x_j}{x_i-x_j}$$

Assumption and boundary condition:

$f(1)=0$ $f(5)=0$ is true and fix
Interpolating the values in the mid

Matlab Code

```
x = [1 2 2.5 3 4 5];
F = [0 5 7 6.5 2 0];


for i = 2:5

    L1 = (x(i)-x(i+1))/(x(i-1)-x(i+1));
    L2 = (x(i)-x(i-1))/(x(i+1)-x(i-1));

    Fr(i) = L1 * F(i-1) + L2 * F(i+1);

    fprintf('Value for Fx at %.2f is %.2f\n', x(i), Fr(i));
end
```

```
>> untitled10
Value for Fx at 2.00 is 4.67
Value for Fx at 2.50 is 5.75
Value for Fx at 3.00 is 5.33
Value for Fx at 4.00 is 3.25
```

Figure 1 Matlab code for first order interpolation

```
x = [1 2 2.5 3 4 5];
f = [0 5 7 6.5 2 0];
for i = 2:5
    if i < 4
        f0 = f(1) * (x(i)-x(i+1)) * (x(i)-x(5)) / ((x(1)-x(i+1)) * (x(1)-x(5)));
        f1 = f(i+1) * (x(i)-x(1)) * (x(i)-x(5)) / ((x(i+1)-x(1)) * (x(i+1)-x(5)));
        f2 = f(5) * (x(i)-x(1)) * (x(i)-x(i+1)) / ((x(5)-x(1)) * (x(5)-x(i+1)));
    else
        f0 = f(1) * (x(i)-x(i-1)) * (x(i)-x(5)) / ((x(1)-x(i-1)) * (x(1)-x(5)));
        f1 = f(i-1) * (x(i)-x(1)) * (x(i)-x(5)) / ((x(i-1)-x(1)) * (x(i-1)-x(5)));
        f2 = f(5) * (x(i)-x(1)) * (x(i)-x(i-1)) / ((x(5)-x(1)) * (x(5)-x(i-1)));
    end
    F = f0 + f1 + f2;
    fprintf('value for fx at %.2f is %.4f\n', x(i), F);
end
```

```
Value for Fx at 2.50 is 5.75
Value for Fx at 3.00 is 5.33
Value for Fx at 4.00 is 3.25
>> untitled10
value for fx at 2.00 is 6.0000
value for fx at 2.50 is 6.8125
value for fx at 3.00 is 6.6667
value for fx at 4.00 is 2.0000
```

Figure 2 Matlab code for second order interpolation

```
clc
x = [1 2 2.5 3 4 5];
f = [0 5 7 6.5 2 0];
for i = 2:5
    if i == 2
        idx = [1 3 4 5];
    elseif i == 3
        idx = [1 2 4 5];
    else
        idx = [1 i-1 i-2 5];
    end
    x0 = x(idx(1)); f0v = f(idx(1));
    x1 = x(idx(2)); f1v = f(idx(2));
    x2 = x(idx(3)); f2v = f(idx(3));
    x3 = x(idx(4)); f3v = f(idx(4));

    L0 = (x(i)-x1)*(x(i)-x2)*(x(i)-x3)/((x0-x1)*(x0-x2)*(x0-x3));
    L1 = (x(i)-x0)*(x(i)-x2)*(x(i)-x3)/((x1-x0)*(x1-x2)*(x1-x3));
    L2 = (x(i)-x0)*(x(i)-x1)*(x(i)-x3)/((x2-x0)*(x2-x1)*(x2-x3));
    L3 = (x(i)-x0)*(x(i)-x1)*(x(i)-x2)/((x3-x0)*(x3-x1)*(x3-x2));
    F = f0v*L0 + f1v*L1 + f2v*L2 + f3v*L3;
    fprintf('value for fx at %.2f is %.4f\n', x(i), F);
end
```

```
Command Window
value for fx at 2.00 is 6.1667
value for fx at 2.50 is 6.3438
value for fx at 3.00 is 7.6667
value for fx at 4.00 is 2.0000
>> Press Ctrl + Shift + P to generate code with Copilot
```

Figure 3 Matlab code for third order interpolation

Result

The function values were estimated using Lagrange interpolation of first, second, and third order. It is observed that as the order of the polynomial increases, the approximation generally improves because more data points are considered. However, accuracy also depends on the proper selection of points. Interpolations using points closer and centered around the required value give better results. Hence, higher-order interpolation does not always guarantee better accuracy unless appropriate points are chosen.